

Lab 7: Radioactive Playground

Benjamin Liou

Aiden Man

Nathan Nguyen

Radioactivity A was plotted against distance r in a linearized graph, plotting $\ln(A)$ against $\ln(r)$. This was conducted for gamma, beta, and alpha decay sources. The gamma decay source, cobalt-60, yielded a slope of -2.03 ± 0.07 and a t' -score of 0.429, verifying the theory that radiation intensity decreases with distance as by the inverse square law. The beta decay source, strontium-90, yielded a slope of -2.37 ± 0.03 and a t' -score of 12.3, indicating strong disagreement with the inverse square law, suggesting potential sources of error. The alpha decay source, polonium-210, yielded a slope of -3 ± 1 that captures the expected value with a t' -score of 1.0. However, the low R^2 value of 0.459 and inconsistency of point locations and vertical error bar sizes relative to the linear fit imply that the source does not follow the inverse square law, reflecting the expected behavior of alpha particles. The radioactive count measurements of cobalt-60 samples dated May 2003 and March 2021 were used to determine the difference in age between the two samples. The difference was calculated to be 4600 ± 300 days, which deviated from the expected value with a t' -score of 6.38, also suggesting potential sources of error.

1. BACKGROUND

Radioactive decay is a process in nuclear physics where atomic nuclei decay into more stable configurations. There are 3 main types of decay: alpha decay, beta decay, and gamma decay. In alpha decay, alpha particles are emitted, as a result of quantum tunneling. These alpha particles consist of two protons and two neutrons, drawing similarities from the helium nuclei. In beta decay (specifically the negative form of beta decay), neutrons are converted to protons, emitting electrons and anti-neutrinos. In gamma decay, high-energy photons, otherwise known as gamma rays, are emitted as high-energy protons or neutrons relax, similar to how electrons in atoms relax and emit photons.

Three different radioactive isotopes correspond to the different forms of radioactive decay: polonium-210 corresponds to alpha decay, strontium-90 corresponds to beta decay, and cobalt-60 corresponds to gamma decay.

Polonium-210 (half-life of 136 days) alpha decays into stable lead-206. Strontium-90 (half-life of 28.8 years) beta decays into yttrium-90 (half-life of 64 hours), which then beta decays into either a grounded or excited state of Zirconium-90

depending on the energy of the electron being emitted. Cobalt-60 (half-life of 5.27 years) beta decays into Nickel 60 which then gamma decays. With three different forms of radiation, a Geiger counter can be used to measure radioactivity by counting the number of ionic particles.

2. THEORY

2A. RADIATION AND DISTANCE:

An inverse square relation between observed radioactivity and distance from the source follows the equation:

$$A = \frac{c}{r^2} = c \cdot r^{-2}$$

where A is radioactivity, r is distance, and c is an arbitrary constant. Linearizing the equation by taking the natural logarithm of both sides simplifies to the following equation:

$$\ln(A) = -2 \cdot \ln(r) + \ln(c)$$

Where the relation forms a line with slope of -2 when plotting $\ln(A)$ against $\ln(r)$. This inverse square relation can be observed for beta and gamma radiation, but will not necessarily be observed in alpha decay. This is because alpha

particles are massive and charged, meaning an increase in distance would introduce an amount of air particles that would shield the detector and greatly reduce the reading from what would be expected.

2B. RADIOACTIVE DATING:

If two radioactive samples are created with the same initial activity, with one at $t = 0$ and the other at $t = T$, where t is time and T is the difference in time between the creation of the two samples, the following equations can be calculated:

$$A_0 = A_1(t = 0) = A_2(t = T)$$

$$A_1(t) = A_0 e^{-lt}$$

$$A_2(t) = A_0 e^{-l(t-T)}$$

$$\frac{A_2}{A_1} = e^{lT}$$

Taking the natural logarithm of both sides of the previous equation and replacing l with the half-life term results in the following:

$$T = \frac{t_{1/2}}{\ln(2)} \cdot \ln\left(\frac{A_2}{A_1}\right)$$

Measuring the radioactivities of the two samples and using 1925.3 days for the half-life of cobalt-60, the number of days between the creation of the samples can thus be determined.

3. EXPERIMENT

Two separate experiments were conducted: one measuring the relation between radioactivity and distance for various radioactive decay types, and one radioactive dating the time difference between the distant creation of two cobalt-60 samples. Before taking any Geiger counter measurements, a background noise measurement was taken and subtracted from each measurement to correct for any ambient radioactivity.

3A. EFFECTS OF DISTANCE

Prior to measurement, each sample was selected such that the time difference between creation and present was less than one half-life in order to allow sufficient radioactive measurement. The following procedure was performed for polonium-210 (alpha decay), strontium-90 (beta decay), and cobalt-60 (gamma decay):

A radioactive sample was placed on the top tray of the sample holder (4cm from the counter) and the data was taken for 60 seconds. The source was then lowered one step further from the detector, each step increasing the distance by 1cm and the data was taken again. This was repeated for each step all the way down the sample holder. If the count dropped below a statistically reasonable number, the time length for the measurement was extended.

3B. AGE OF CO-60 SAMPLES

Poisson statistics state that the standard error in the count measurement is equal to the square root of the actual measurement. Thus, to reduce proportional error, the time interval was extended to 300 seconds, the approximate length of one Clash Royale match (considering overtime). This would result in greater counts and proportionally smaller errors. Selecting two samples of cobalt-60 with statistically significant time difference between creation dates, both samples were measured at the top tray of the sample holder. This specific experiment selected one sample created in May 2003, and another in March 2021.

4. DATA, RESULTS, AND ANALYSIS

For the experiment measuring the effects of distance on radioactivity, the linearized data with $\ln(A)$ against $\ln(r)$ was plotted. The slopes for each isotope are -2.37 ± 0.03 for strontium-90, -2.03 ± 0.07 for cobalt-60, and -3 ± 1 for polonium-210. With an expected slope of -2 for the inverse square relation, the t' -scores are thus 12.3, 0.429, and 1.0 respectively. The t' -score for

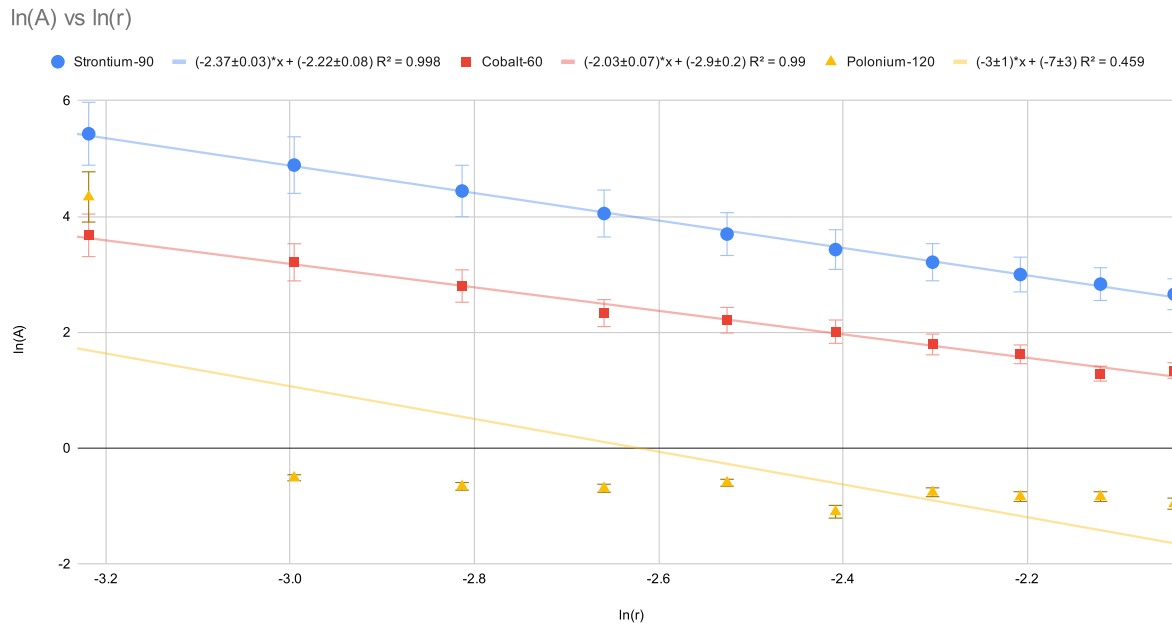


Figure 1: Natural logarithm of radioactivity vs. natural logarithm of distance for strontium-90, cobalt-60, and polonium-210. The solid line represents the linear best fit to the experimental data points. The equation of the fit and the R^2 value are shown on the graph.

strontium indicates a strong disagreement with the expected value, with the t' -score of 12.3 being far greater than 3. The t' -score for cobalt indicates an agreement within the uncertainty, with 0.429 being less than 1. For polonium, although the error captures the expected value and yields a t' -score of 1.0, the graph reveals that a non-power fit would likely relate radioactivity and distance better. The error bar of every data point on the graph does not capture the line, and the R^2 has a low value of 0.459, supporting the distinction between this data and an inverse square relation. This is consistent with the theory that polonium-210, exhibiting alpha decay, would not follow the same inverse square relation as strontium-90 or cobalt-60, which each exhibit beta and gamma decay respectively.

For cobalt dating, Poisson statistics indicate that the uncertainty in the count is equal to the square root. Considering this, the radioactivity readings for the May 2003 sample read 2300 ± 50 with approximately 2.17% error, and for the March 2021 sample 11900 ± 100 with

approximately 0.840% error. Considering error propagation with a 5% systematic error in each sample and the Poisson statistic uncertainty, the percent error gets compounded to approximately 7.44%. Thus, performing the calculation and applying error propagation yields a time difference of 4600 ± 300 days between the two samples. Comparing this to the expected value of approximately 6514 days, determined by calculating the difference between the May 1st, 2003 and March 1st, 2021, the t' -score yields 6.38, signifying disagreement between the data and the expected value.

5. CONCLUSION

The t' -score of cobalt-60 shows agreement within the uncertainty, verifying the inverse square relation between observed radioactivity and distance from the source for gamma particles.

However, the t' -score of strontium-90 signifies disagreement with the inverse square relation for beta particles, when it is expected to be in agreement. This could be due to detector miscalibration after years of laboratory use. For polonium-210, its non-linear graph indicates a poor fit to the inverse square relation, which is consistent with the behavior of alpha particles.

The t' -score from cobalt-60 dating analysis shows significant disagreement between the sample's estimated age and actual age. This could be due to detector miscalibration, resulting in an underestimation of the newer sample's age with respect to the older sample.

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